

libtt: An Open-Source TPU-Style XLA Stack for Tenstorrent Accelerators

libtt Technical Report

libtt Project

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Abstract

Designing a high-performance tensor accelerator is difficult. Building the software stack that makes it broadly useful is harder still. Such a stack must run existing open-source machine-learning systems across models and workloads while mapping logical tensors onto tiled device memory, topology-aware communication, and specialized kernels. It must expose enough of the hardware for optimization without requiring each framework, serving engine, or model to be ported by hand. libtt adopts the TPU software boundary for Tenstorrent. StableHLO and PJRT let it reuse software from the JAX ecosystem above the compiler rather than porting each framework and model, while MLIR supplies mature infrastructure for lowering logical tensors onto tiled memory and topology-aware execution. The same XLA boundary provides an architectural path to PyTorch through TorchTPU. libtt packages the open-source compiler, runtime, device kernels, and runtime assets in one shared library. We demonstrate it with upstream SGLang-JAX and Qwen3-8B. A sequence of graph, layout, runtime, and kernel optimizations raises measured 128-token end-to-end generation from 16.265 to 27.619 tokens/s, a 69.81% improvement, and delivers 27.753 decode tokens/s, 11.51% faster than the measured tt-inference-server reference on the same Blackhole P150 and workload. The result shows that a stable XLA boundary can keep the upper software stack close to upstream while concentrating hardware-specific performance work in the compiler and lower runtime.

1 Introduction

This report asks whether Tenstorrent accelerators can support programs from the XLA ecosystem through a standard compiler boundary—without a device-specific model or serving-engine fork—while retaining hardware-specific performance. libtt’s answer is an open-source, self-contained XLA stack built around PJRT [6] and StableHLO [7]. It compiles and executes framework programs while keeping Tenstorrent-specific code below that boundary. The source code, patches, and build files are available in the public libtt GitHub repository [35].

The implementation evaluated here uses JAX. Upstream SGLang-JAX retains its tokenizer, scheduler, Qwen model, and paged KV cache [9], while libtt compiles and executes the exported StableHLO. JAX is the demonstrated frontend, not the architectural boundary. Google’s announced TorchTPU design captures PyTorch graphs with Dynamo, translates PyTorch operations to StableHLO, and uses XLA as its primary backend compiler [2]. This gives libtt a path to support PyTorch as that interface becomes available. The required adapter and operation coverage have not yet been implemented or measured.

The XLA boundary has two practical advantages. First, framework and serving code can remain close to upstream instead of being replaced by a Tenstorrent-specific model and inference engine. Second, performance work can reside in TT-MLIR [23], TTNN [26], TT-Metalium [21], and device kernels, where graph structure, layout, precision, and data movement remain visible. The same boundary can carry inference or training graphs; this report measures inference only.

The report makes four contributions. First, it describes a single `libtt.so` artifact containing the Tenstorrent PJRT backend, compiler, runtime, kernels, and runtime assets. Second, it demonstrates this XLA boundary with upstream SGLang-JAX and identifies the future PyTorch/TorchTPU path. Third, it presents compiler and kernel optimizations that cross StableHLO, TT-MLIR, TTNN, and TT-Metal without modifying the SGLang scheduler or Qwen model [31]. Finally, it measures a 69.81% end-to-end improvement from 16.265 to 27.619

tokens/s and 27.753 decode tokens/s, 11.51% above the matched tt-inference-server v0.10.0 reference [20].

The following sections describe the system boundary, compiler and runtime representations, individual optimizations, and measured effects.

2 Tenstorrent hardware and software stack

The performance comparison later in this report involves more than two kernel libraries. The reference TTIS path includes a serving engine, a Tenstorrent-specific engine backend, a model-specific implementation, TTNN, TT-Metalium, the user-mode driver, firmware, and the device. This section separates those layers and identifies the code required to join them.

2.1 Blackhole and the Tensix execution model

The benchmark uses a Blackhole P150. The Blackhole processor exposed by current P150 firmware has 120 Tensix cores, 180 MB of SRAM (1.5 MB per core), and up to 32 GB of GDDR6 memory [12]. The host reaches the card over PCIe. Device DRAM is distinct from host memory, and the SRAM local to a Tensix core is explicitly managed working storage rather than a hardware cache.

Tensix cores communicate through a two-dimensional network on chip. A typical TT-Metalium program assigns a reader data-movement kernel, a compute kernel, and a writer data-movement kernel to each participating core. The three kernels exchange tiles through circular buffers in local SRAM and can overlap memory traffic with arithmetic [21, 22]. Matrix and vector results pass through the unpacker, SrcA/SrcB and Dst registers, and the packer before returning to L1.

These details matter for batch-one decode. The weights are much larger than a token activation, so every layer repeatedly streams weights from GDDR6. Core count, NoC multicast, circular-buffer capacity, and the number of times an intermediate is packed or written to DRAM directly determine token latency.

2.2 Why the TPU software architecture fits Tenstorrent

TPUs and Tenstorrent accelerators have two important things in common. Their main matrix paths use tiled tensors in device DRAM, and their systems connect chips with fast direct links. Both features make compiler support central to performance.

First, the physical layout of a tensor matters across operations. XLA assigns tiled layouts to TPU buffers and pads shapes that do not fill a tile [8]. Tenstorrent uses 32-by-32 tiles for most Tensix matrix compute and data movement. Tiled tensors can be stored in DRAM or L1 [18, 27]. The compiler and runtime must choose the tile layout, padding, memory location, and sharding, then keep compatible layouts between producers and consumers. If every operation changes the layout, the extra reads, writes, and conversions can remove much of the accelerator’s benefit.

This makes a compiler a basic part of a fast TPU or Tenstorrent path. It must plan physical layouts across the graph, not only select an arithmetic kernel. The optimizations in this report follow that model: StableHLO keeps logical tensors above the device boundary, while TT-MLIR and TTNN carry the tiled physical layout below it.

The common GPU path is different. GPU tensors can stay row-major or strided in global memory. A CUDA kernel can read that array directly and load tiles into registers or shared memory for its own work [5]. The next kernel can read the same row-major or strided buffer. It does not first need a persistent tiled DRAM layout. This is why eager GPU execution can be efficient: each operation can launch an optimized kernel on the existing data without a layout conversion before every launch. GPU graph compilers still improve fusion, launch overhead, and scheduling, but a whole-graph compiler is less central to the basic execution model.

Second, TPU and Tenstorrent systems both provide fast links between chips. TPU slices connect chips through high-speed ICI in two- or three-dimensional topologies [3]. A Blackhole P150 card has four 800 Gbps ports that connect directly to other Blackhole cards [12]. Tenstorrent Galaxy systems connect Blackhole chips in grid or torus topologies [34]. These links let the system act as a mesh of accelerators rather than a set of unrelated PCIe devices.

At this scale, the compiler and runtime must decide how to shard tensors, place work, and schedule collectives over the chip topology. This report benchmarks one P150, so it does not measure multi-chip execution. The

interconnect still supports the same long-term software design: graph compilation above a device mesh with explicit layout, placement, and communication.

Execution path	Device data and links	Software consequence
XLA on TPU	Tiled buffers in device memory; chips linked by high-speed ICI.	The compiler plans layout, sharding, communication, and execution.
libtt on Tenstorrent	The main matrix path uses tiles in DRAM and L1; Blackhole cards have direct high-speed links.	Keep layouts across operations and plan work over the device mesh.
Eager GPU path	Row-major or strided buffers can remain in global memory; kernels tile on load.	Optimized kernels can run directly on existing buffers without a graph-wide layout plan.

This is why the TPU software architecture fits Tenstorrent. Both need software that starts from a graph, assigns tiled physical layouts, and plans execution over a connected set of chips. A GPU-style eager interface can also work, but it leaves more layout and communication decisions to model and operator code.

2.3 Runtime and operator layers

The software stack exposes the device at several levels:

Layer	Responsibility
TT-UMD	Discovers devices and provides the user-mode PCIe, memory, and command transport used by higher runtimes [29].
TT-Metalium	Allocates device resources, builds command programs, and compiles and dispatches reader, compute, and writer kernels [21, 22].
TTNN	Provides Python and C++ tensor operations, tensor layouts, memory configurations, sharding, program selection, and tracing on top of TT-Metalium [26].
TT-Transformers	Implements Transformer structure, model configuration, paged attention, KV-cache behavior, prefill/decode paths, and device-specific layouts with TTNN calls [28].
TT vLLM backend and TTIS	Connects model implementations to the serving scheduler, request batching, cache allocation, model loading, and OpenAI-compatible service [15, 20].

TTNN is therefore an operator library and direct runtime interface, not a framework compiler boundary by itself. It exposes preoptimized operations and selects TT-Metalium programs, but a caller using TTNN directly still constructs the model, chooses layouts and shapes, manages state, and determines when host and device tensors cross the boundary.

2.4 How TTIS reaches TTNN

Tenstorrent uses vLLM for scheduling, continuous batching, paged attention, and the OpenAI-compatible server. Its current integration guide describes a TT platform backend/plugin and a maintained vLLM fork [15]. The backend supplies platform configuration, model registration, model loading, worker and device initialization, KV-cache management, input preparation, execution, and TT engine behavior. The plugin reduces the amount of code that must live directly inside a vLLM fork, but these Tenstorrent-specific engine responsibilities remain.

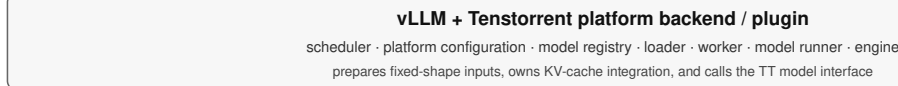
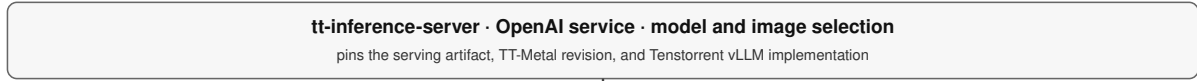
Adding a model requires more than registering a generic Hugging Face class. The documented integration contract requires:

1. a TTNN paged-attention implementation using the TT cache-fill, cache-update, and paged decode operations;
2. a TT-Transformers generator with initialization, KV-cache allocation, prefill, decode, trace warm-up, output, and capability interfaces;
3. fixed-shape padding and trace behavior compatible with the TTNN execution path;
4. model registration and configuration in the TT vLLM backend; and
5. for new model types or inputs, possible changes to the platform, loader, worker, model runner, or engine [15].

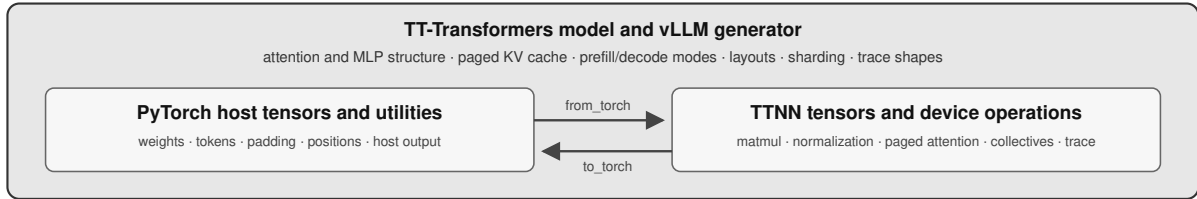
The v0.10.0 artifact measured in this report pinned TT-Metal commit `e867533` and Tenstorrent vLLM commit `22be241`. In that release, the TT platform code was carried in the Tenstorrent vLLM path. Current upstream code is increasingly packaged as a plugin, but the model-facing interface and TT-specific execution logic described above remain. We distinguish this packaging change from the v0.10.0 performance measurement [20].

Reference TTIS path and its explicit integration layers

SERVING AND ENGINE



MODEL-SPECIFIC PORT



RUNTIME, KERNELS, AND HARDWARE

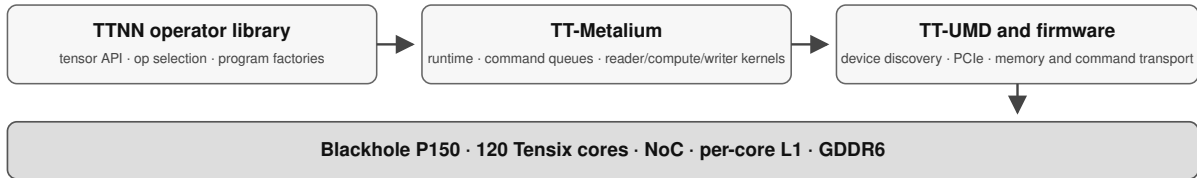


Figure 1: The reference TTIS path from serving through the model port and lower runtime.

2.5 PyTorch and TTNN in the same model implementation

TT-Transformers intentionally uses PyTorch and TTNN in the same Python model. The Qwen/Llama model source imports both libraries. Host tokens, positions, padding, checkpoint values, and some output processing use `torch.Tensor`, while `ttnn.from_torch` creates TTNN tensors for device execution and `ttnn.to_torch` returns selected outputs to the host [28]. The vLLM generator similarly accepts PyTorch token and page-table tensors, allocates the paged KV cache as TTNN tensors, and calls model-specific prefill and decode methods.

This does not mean that the main Transformer matmuls execute in PyTorch on the CPU. Those operations use TTNN and TT-Metalium. It does mean that the reference path contains a hand-written, Tenstorrent-specific upper model layer that decides where the PyTorch/TTNN conversions occur and encodes padding, trace shapes, cache layout, dtype, sharding, collectives, and model-specific fast paths. Supporting a new architecture requires extending this model port and its serving adapter.

2.6 Contrast with the libtt boundary

Concern	TTIS and TT-Transformers path	libtt path
Serving engine	vLLM with a TT platform backend/plugin	The measured SGLang-JAX engine remains above PJRT
Model	TT-Transformers implements a TT-specific Transformer and generator	The measured upstream JAX Qwen model is exported through XLA to StableHLO
Framework/device boundary	Python generator methods and explicit PyTorch/TTNN tensor conversion	XLA’s StableHLO and PJRT interfaces
Paged KV cache	Coordinated by vLLM backend code and TT-Transformers TTNN operations	Expressed by SGLang-JAX and lowered through TT-XLA/TT-MLIR
Performance work	TT-specific engine, model, TTNN, program, and kernel code	TT-MLIR transformations, TTNN runtime policy, and TT-Metalium kernels
Deployment	Versioned server images pin compatible TT-Metal and vLLM revisions	One <code>libtt.so</code> pins and contains the compiler and runtime stack

The direct TTNN approach gives the model author explicit control at every layer. libtt instead preserves a stable compiler boundary and attempts to recover the same hardware-specific performance below it. The comparison in this report tests whether that lower-stack approach can keep an XLA frontend close to upstream without giving up decode performance.

3 System organization

3.1 Deployment and frontend boundary

PJRT lets XLA frontends call device-specific compiler and runtime plugins through one interface [6]. libtt implements that interface for Tenstorrent:

XLA frontend process (JAX in the measured path)

```
-- dynamically loads libtt.so through PJRT
  |-- compiles StableHLO for Tenstorrent
  |-- allocates and transfers device buffers
  |-- loads and executes serialized TTNN programs
-- initializes the embedded TT-Metal runtime assets
```

libtt’s C++ glue extracts the embedded archive to a fingerprinted temporary directory, sets `TT_METAL_RUNTIME_ROOT` before PJRT starts, links the upstream plugin, and exports only the PJRT entry points. Most code comes from pinned open-source dependencies and the applied patch series.

Today, libtt is a set of patches on pinned versions of TT-XLA, TT-MLIR, and TT-Metal/TTNN. The patches cover the build, compiler passes, runtime choices, and device kernels. Keeping the patches in one repository makes them easy to compare with upstream code and lets one change span all three projects.

The compiler and TT-Metal runtime do not need to be installed on the target. With compatible drivers and firmware, `libtt.so` contains the software needed to compile and run StableHLO programs. This is the intended similarity to `libtpu.so` [11].

libtt treats PJRT and StableHLO as the integration boundary. In the measured path, SGLang-JAX keeps its model, tokenizer, scheduler, paged KV cache, and inference engine. The Tenstorrent-specific work stays below that boundary instead of requiring a forked Qwen implementation or a rewritten serving layer. A future PyTorch path can reach the same boundary through TorchTPU once the backend interface and required operation coverage are available [2].

The SGLang-JAX prototype uses a small TT integration patch. It adds the TT attention backend, device selection, weight annotations and loading, and KV-cache support [33]. The scheduler, tokenizer, and Qwen

libtt: one XLA/PJRT artifact from StableHLO to Tensix execution

XLA FRONTENDS · OUTSIDE LIBTT · LARGELY UNCHANGED

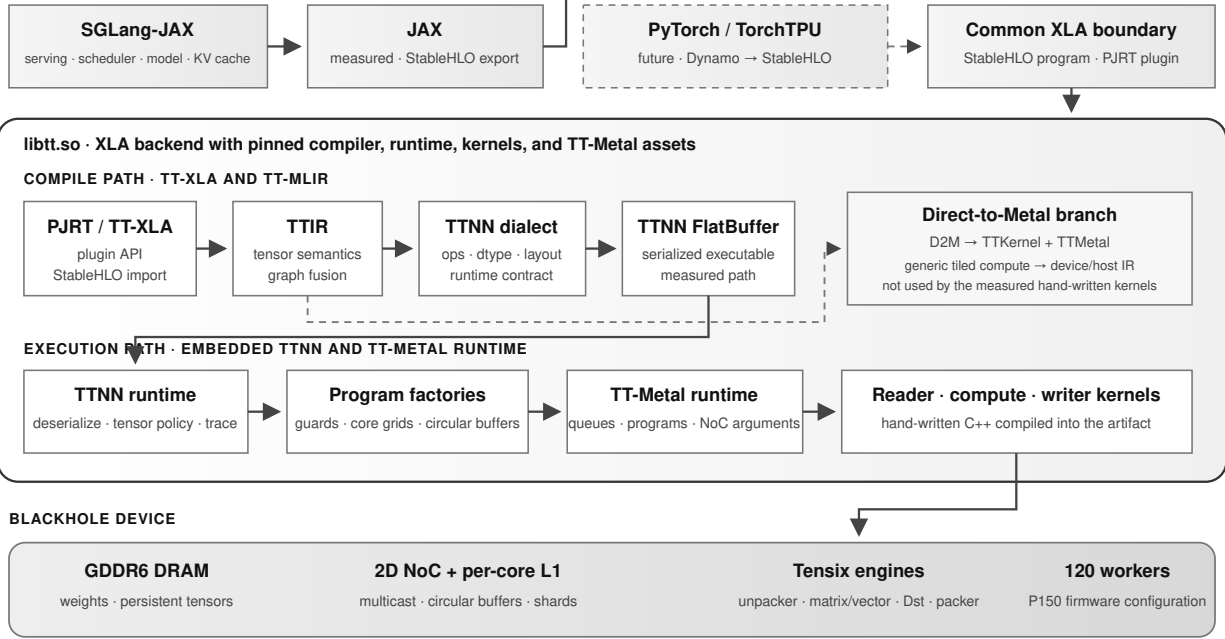


Figure 2: libtt XLA boundary, compilation, runtime, and device layers.

model structure stay close to upstream. Most of the JAX software does not change. It only needs a small device integration layer, not a new framework, serving engine, or Transformer implementation.

This provides two advantages:

- XLA frontend code can stay upstream, as demonstrated here with SGLang-JAX; and
- performance can be added in TT-MLIR, TTNN, TT-Metal, and device kernels, where layout, fusion, precision, and data movement are visible.

We evaluate inference, but PJRT and StableHLO are neither framework- nor inference-specific. JAX or PyTorch training support can use the same boundary after the required operations, collective behavior, optimizer graphs, and numerical validation are available.

3.2 Integrated build

libtt uses one Bazel build for the whole stack. Bazel pins TT-UMD [29], SFPI [17], TT-Metal, LLVM, StableHLO, TT-XLA [30], TT-MLIR, Shardy, and supporting C++ libraries [1]. It applies the libtt patches and adds build targets where needed. The `//:tt` target builds the compiler, runtime, kernels, and runtime assets, then links them into `bazel-bin/libtt.so`.

This organization has three practical effects:

- Source revisions, overlays, patches, toolchains, and link rules are inputs to one build.
- Compiler, schema, runtime, program-factory, and kernel changes can be tested together.
- An agent can inspect and edit every layer in one workspace, rebuild one target, and run one benchmark.

Tenstorrent’s compiler overview describes the same open path from framework frontends through TT-MLIR dialects to TT-Metalium [19, 24]. libtt builds a pinned part of that path as one artifact.

3.3 Cross-layer optimization

The matmul-SwiGLU epilogue is an example of a cross-cutting optimization. It cannot be implemented only as a graph rewrite or only as a device kernel. It needs coordinated changes across six layers:

1. A graph pass must prove that a paired gate/up matmul feeds the exact `silu(gate) * up` expression.
2. TTNN IR needs an operation-level representation that preserves the fused semantic through lowering.

3. The serialized executable needs to carry the new `matmul` attribute.
4. The runtime must dispatch that operation into TTNN.
5. TTNN must select a program factory only for supported shapes, dtypes, layouts, and hardware.
6. Reader, compute, and writer kernels must keep the paired tiles resident in Dst and emit only the half-width final result.

The full stack is open source, so libtt can change all of these layers and test them together in one build. This is what makes cross-cutting optimizations such as `matmul-SwiGLU` practical. The six changes form one patch concept even though they touch several abstraction levels and two upstream repositories.

3.4 TT-MLIR IR levels

MLIR supports dialects at different abstraction levels [4]. TT-MLIR uses them for model semantics, layouts, tiled computation, device kernels, and host/device control [24]. Each optimization should use the highest level that still has the information it needs.

Table 4: The TT-MLIR IR ladder and where libtt’s measured path uses it.

Representation	Abstraction and purpose	Relationship to current libtt optimizations
StableHLO	Framework-portable tensor graph with specified operation semantics [7].	Input to TT-XLA/TT-MLIR. Framework algebra such as expanded SiLU is visible here and after import, but libtt’s patches generally match it in TTIR/TTNN passes.
TTIR	Hardware-aware, high-level tensor IR. It preserves named tensor operations while permitting fusion and canonicalization before a concrete runtime API is chosen [24].	JAX RMSNorm recognition, expanded-SiLU recovery, shared-LHS matmul grouping, role propagation, and part of QKV/RoPE fusion.
TTNN dialect	High-level tensor IR designed to model the TTNN library closely. Its types and attributes carry tiled layouts, memory spaces, sharding, and device data types [24, 25].	QKV composite fusion, <code>matmul_swiglu</code> , layout selection, cache return types, and lowering to the serialized TTNN operation graph.
TTNN FlatBuffer	Serialized executable operation graph consumed by the TTNN runtime. It is an artifact rather than an MLIR dialect.	Carries operation parameters such as the fused-SwiGLU matmul flag across the compiler/runtime boundary.
D2M dialect	Generic tensor/memref computation analogous to <code>linalg.generic</code> , augmented with grids, sharded tensors, circular buffers, and explicit data movement [24].	An alternative direct-to-metal route. The measured custom SwiGLU and down-projection kernels are hand-written TT-Metal/TTNN patches, so they do not pass through D2M.
TTKernel dialect	Low-level device-kernel IR exposing circular buffers, tile registers, NoC transactions, and synchronization with an intended near one-to-one mapping to TT-Metal kernels [24].	Relevant to generated direct-to-metal kernels; not the representation of the hand-written C++ kernels measured here.

Representation	Abstraction and purpose	Relationship to current libtt optimizations
TTMetal dialect	Host/device interop IR for allocation, transfers, program creation, and enqueue operations [24].	Part of the direct-to-metal compiler route. The measured TTNN route instead invokes TT-Metal host APIs from the TTNN runtime.

The main lowering branch in this report is therefore:

`StableHLO → TTIR → TTNN dialect → TTNN FlatBuffer`

↓

`TTNN C++ runtime`

↓

`TT-Metal host + C++ kernels`

TT-MLIR also supports the lower-level branch:

`TTIR → D2M → TTKernel + TTMetal → generated device/host program`

Only the first branch runs the benchmarked patches. Their hand-written C++ program factories and kernels are not TTKernel IR optimizations.

3.5 TTNN runtime and the serialized executable

TTNN IR models operations such as matmul, RMSNorm, SDPA, and reshape. Lowering serializes them into a FlatBuffer. The runtime deserializes the graph, creates tensors, sets memory configurations, validates each operation, and selects a program factory. The RMSNorm patch uses the concrete input shape here to choose a 16-core width-sharded layout; the compiler still emits a normal RMSNorm.

3.6 TT-Metal programs and Tensix kernels

TT-Metalium uses cooperative dataflow rather than a GPU thread hierarchy [22]. A program usually has:

- a reader data-movement kernel to fetch tiles from DRAM or another core into L1 circular buffers;
- a compute kernel to unpack tiles, drive matrix/vector engines, and create result tiles; and
- a writer data-movement kernel to drain result buffers to their destination.

Circular buffers are bounded queues shared by the threads of a Tensix core [13]. Reader, compute, and writer kernels can overlap on different tiles when their buffer and semaphore protocols are correct.

The compute engine exposes SrcA, SrcB, and Dst register sets. Matrix results land in Dst; vector operations can transform them; the packer writes them to an L1 circular buffer [14]. With 16-bit Dst storage and double buffering enabled, the active half contains eight 32-by-32 tiles. That capacity determines the successful SwiGLU geometry: four gate and four up tiles fill Dst exactly.

3.7 Tiles, memory, and low-precision weights

TTNN’s standard tile is 32 by 32 elements with four 16-by-16 faces [18]. Final dimensions are padded, so batch-one decode still presents a full tile row. Kernels must account for both the logical and padded shapes.

Large weights reside in device DRAM; L1 is smaller, faster, and private to each worker. TTNN `MemoryConfig` values describe interleaved or sharded storage and DRAM or L1 placement. TT-MLIR layout attributes similarly encode how a logical tensor maps to devices, cores, physical shards, memory space, and padding [25].

We use BF16 activations and outputs with BFLOAT8_B weights. BFLOAT8_B is block floating point: 16 values share an exponent [27]. It reduces the dominant weight traffic in batch-one decode, with a precision and packing trade-off. The custom matmuls use BF16 packer-L1 accumulation: partial sums are packed into an L1 slot and reloaded for later K blocks, leaving Dst available for the current block’s tile group.

3.8 Decode bottlenecks

Qwen3 is a decoder-only Transformer family [31]. A simplified layer is:

$$\begin{aligned}
u &= \text{RMSNorm}(h), \\
h' &= h + \text{Attention}(Q(u), K(u), V(u); K_{\text{cache}}, V_{\text{cache}}), \\
v &= \text{RMSNorm}(h'), \\
g &= vW_{\text{gate}}, \quad r = vW_{\text{up}}, \\
h_{\text{next}} &= h' + (\text{SiLU}(g) \odot r)W_{\text{down}}.
\end{aligned}$$

RMSNorm and SwiGLU are standard model concepts [10, 32]. Prefill shares each weight read across many rows. Serial decode applies a logical $1 \times K$ by $K \times N$ projection for every token, so weight traffic and intermediate materialization dominate. The main optimizations add width parallelism, reduce DRAM traffic, or avoid packing an intermediate from Dst.

4 Optimization of the model path

We group the changes by engineering concept and implementation level rather than by commit. Git revisions remain in the data files.

Table 5: Current model-path optimization concepts, implementation levels, and measured effects.

Concept	Principal level / IR	Measured effect
Recover JAX RMSNorm	TTIR graph rewrite	+19.28% marginal foundation effect
Lower SiLU calls	StableHLO-to-TTIR rewrite	No positive decode effect detected
Recognize expanded SiLU	TTIR graph rewrite	+3.30%
Fuse rank-3 decode RoPE	TTIR/TTNN composite resolution	+2.89% marginal foundation effect
Fuse Qwen decode QKV structure	TTIR ordering + TTNN fusion	+1.15%
Preserve KV-cache result types	TTNN type inference	+0.54% marginal; near measured drift
Enable BF8 activation lowering	Compile options and TTNN lowering	+0.26% marginal; near measured drift
Permit decode layouts	TTNN/TT-Metal validation	No positive decode effect detected
Width-shard decode RMSNorm	TTNN runtime policy	+13.61%
Combine two shared-LHS matmuls	TTIR fusion	-1.69%; enables epilogue
True matmul-SwiGLU epilogue	TTNN IR/FlatBuffer/runtime + TT-Metal	+2.73%
Generalize and remove fallback	TTNN matching/runtime simplification	-0.19%
Sweep SwiGLU K blocking	TT-Metal program and CB geometry	+0.27%
Specialize down projection	TTNN program selection + TT-Metal	+5.96% pure decode
Trace one-tile prefill	TTNN/TT-Metal runtime trace	-90.44% TTFT; +10.23% E2E

Patch index:

RMSNorm graph	tt_mlir_fuse_jax_rms_norm.patch
SiLU call	tt_mlir_lower_silu_call.patch
expanded SiLU	tt_mlir_fuse_expanded_silu.patch
decode RoPE	tt_mlir_fuse_rank3_rope_decode.patch
QKV structure	tt_mlir_qwen_decode_qkv_projection_fusion.patch
KV-cache types	tt_mlir_kv_cache_dtype_return_types.patch
BF8 path	tt_mlir_single_chip_activation_dtype_lowering.patch
	fast_bfloat16_bfp8_pack.patch
decode layouts	sdpa_decode_allow_1i_interleaved_q.patch
	layernorm_allow_single_core_height_sharded.patch
RMSNorm sharding	tt_mlir_sharded_decode_rms_norm.patch
shared-LHS pair	tt_mlir_fuse_shared_lhs_matmul_pairs.patch
SwiGLU vertical	tt_mlir_fuse_matmul_swiglu.patch
	matmul_swiglu_epilogue.patch
down projection	down_projection_110_core.patch

The original foundation measurement combined these patches into one +22.58% stage. A new leave-one-concept-out experiment below separates their marginal effects and measures a fresh total gain of +24.03%. The effects overlap and do not add to the total. The blocking and down-projection tests use a direct streaming clock. Their final configuration forms the last stage of the cumulative sequence using streaming end-to-end throughput; the same-build A/B tests below isolate the effects of the two kernel changes.

The first group recovers model semantics in TTIR and TTNN before lowerings discard the algebra and graph roles needed to recognize them.

4.1 RMSNorm recognition

JAX can lower RMSNorm to square, reduce, epsilon addition, reciprocal square root, scaling, and reshapes. A TTIR pattern proves this structure and replaces it with one RMSNorm operation while the algebra and use-def graph are still available. TTNN can then apply its normalization and layout logic. Removing this patch from the otherwise complete foundation lowers throughput from 19.894 to 16.678 tokens/s. Restoring it gives a +19.28% marginal speedup. It also enables the later runtime-sharding patch.

4.2 SiLU call lowering

Some JAX graphs contain a function call named `silu`. The foundation patch replaces that call with a TTIR SiLU operation during StableHLO legalization. Removing the patch changes throughput by +0.45%, so this exact graph does not show a positive decode effect. The patch remains useful for graph coverage.

4.3 Expanded-SiLU recognition

The graph represents

$$\text{SiLU}(x) = x \sigma(x) = \frac{x}{1 + \exp(-x)}$$

with casts, broadcasts, reshapes, and a splatted scalar one. This is separate from the foundation call-lowering patch. The later `tt_mlir_fuse_expanded_silu.patch` looks through view-like operations, verifies the constant and shared input, and replaces the tree with SiLU. The isolated gain is +3.30%.

4.4 Rank-3 RoPE and QKV projection structure

Qwen projects Q, K, and V together, applies per-head RMSNorm to Q and K, reshapes them by head count, and applies rotary position embedding. Two patches recover this structure:

- `tt_mlir_fuse_rank3_rope_decode.patch` accepts JAX’s rank-3 decode form, temporarily maps it to the existing rank-4 composite, and restores the expected result shape.
- `tt_mlir_qwen_decode_qkv_projection_fusion.patch` orders projection roles, validates contiguous Q/K/V bounds, head counts, and head dimensions, and replaces materialized slices/reshapes with `nlp_create_qkv_heads_decode`. Q and K RMSNorm are recreated on the fused rank-4 results.

TTIR supplies candidate ordering and role information; TTNN owns the composite operation and runtime contract. QKV fusion adds +1.15%. Removing rank-3 RoPE fusion from the foundation lowers throughput by 2.81%; restoring it gives a +2.89% marginal speedup.

4.5 KV-cache dtype and role propagation

KV-cache update operations are stateful boundaries. Their return types must carry the selected device dtype, and graph-role metadata must survive unary operations so later fusions can still identify query, key, and value paths. The TT-MLIR and TT-XLA patches keep the low-precision fused graph typed and recognizable. Removing KV-cache return-type preservation lowers throughput by 0.54%; restoring it gives a +0.54% marginal speedup.

The next group selects precision and layout after the graph operations have been recovered.

4.6 BF8 lowering and fast host packing

We request `bfp_bf8` for eligible weights. TT-XLA enables single-chip lowering, TT-MLIR assigns the device dtype, and TTNN/TT-Metal consume BFLOAT8_B tiles. `fast_bfloat16_bfp8_pack.patch` speeds conversion of BF16 host weights to 32-by-32 BFP8 tiles.

These changes attack two different phases:

- BF8 storage reduces steady-state device weight traffic during decode.
- faster packing reduces model-load and preparation cost on the host.

BF8 weights and fast host packing already existed in the functional baseline. The foundation adds single-chip activation lowering through one TT-XLA patch and one TT-MLIR patch. Removing those two patches together lowers throughput by 0.25%; restoring them gives a +0.26% marginal speedup.

4.7 Width-sharded decode RMSNorm

The default logical height of RMSNorm during batch-one decode is one. A generic height-parallel program cannot occupy many cores. At run time, we recognize a device-resident, tiled, unsharded input with sequence length one and width at least 2048. When the width divides into 16 tile-aligned shards, it:

1. creates an 8-by-2 row-major core grid;
2. width-shards the tensor into the cores' L1 memories;
3. derives the layernorm program configuration from that shard specification;
4. executes sharded RMSNorm; and
5. restores the requested output memory configuration.

The two conversions cost less than the width-parallel reduction saves. The isolated gain is +13.61%. This is runtime policy, not a new IR operation: the compiler identifies RMSNorm, while the runtime knows the tensor and device geometry.

4.8 Decode-specific validation changes

Two small TT-Metal patches admit layouts selected by the optimized graph:

- SDPA decode accepts an L1-interleaved query rather than requiring the prior sharded form.
- layernorm accepts the single-core height-sharded case produced on the decode path.

These narrow changes keep shape and layout checks while admitting layouts used by other optimized paths. Removing both changes throughput by +0.66% in this experiment, so no positive decode effect is detected for the measured graph. Their role is compatibility rather than the source of the foundation speedup.

The MLP experiments test which intermediate and data-movement boundaries must be removed for fusion to improve decode throughput.

4.9 Shared-LHS gate/up projection

Qwen's two first MLP projections share an activation:

$$xW_{gate}, xW_{up} \longrightarrow x[W_{gate} \ W_{up}].$$

`tt_mlr_fuse_shared_lhs_matmul_pairs.patch` changes TTIR fusion eligibility so a pair can be combined. This removes duplicate activation reads and launches, but writes a doubled-width output that SwiGLU slices and rereads. Throughput changes by -1.69%. The representation becomes useful when the epilogue consumes it without materializing the combined output.

4.10 Consumer-side SiLU fusion

An intermediate experiment placed SiLU as a unary activation on the following TTNN multiply. That removes one elementwise operation but retains the expensive boundary:

```
combined matmul → pack full gate/up tensor → memory
                  → read/unpack → SiLU + multiply → pack result
```

Throughput changes by -0.65%. Removing an operation does not help when the same large intermediate is still written and read.

4.11 Dst-resident matmul-SwiGLU epilogue

Two patches implement the producer-side epilogue:

- `tt_mlr_fuse_matmul_swiglu.patch` recognizes the TTNN graph, introduces the fused matmul semantic, extends the FlatBuffer representation, serializes it, and invokes the corresponding runtime path.
- `matmul_swiglu_epilogue.patch` adds TTNN validation, a program factory, and the TT-Metal reader/receiver/sender/compute kernels.

The contract supports one physical tile row, BF16 activation/output, BFLOAT8_B weights, DRAM-interleaved tensors, no transpose or bias, and a grid that can supply 96 workers. For Qwen3-8B, each worker owns four gate tiles and four corresponding up tiles—exactly the eight active 16-bit Dst slots.

The program performs the following sequence:

1. two sender cores multicast activation K blocks over the NoC;

2. each of 96 workers reads its paired gate/up weight tiles;
3. the matrix engine accumulates all eight output tiles;
4. partial K results use BF16 packer-L1 accumulation;
5. final gate/up partials are loaded together into Dst;
6. SiLU transforms Dst slots 0–3;
7. each gate tile multiplies its corresponding up tile in place; and
8. only four final BF16 tiles are packed and written.

The doubled-width gate/up tensor is never materialized. End-to-end throughput improves by +2.73%.

4.12 Prefill coverage and fallback removal

We then generalized the matcher from decode-specific naming to any supported one-tile-row input, which includes short prefill. The older consumer-side SiLU-multiply fallback was removed; validation remains at the TTNN/TT-Metal boundary, where unsupported dtype, layout, shape, and hardware combinations fail explicitly.

Throughput changes by -0.19%. The change removes code and adds prefill coverage without changing the fast kernel or completion hash.

4.13 SwiGLU K-blocking sweep

The initial epilogue uses `in0_block_w = 2`. Every two K tiles it packs eight partial tiles into L1 and later reloads them. Larger blocks reduce partial traffic but need larger activation and weight circular buffers.

`matmul_swiglu_epilogue.patch` now exposes `TT_METAL_SWIGLU_IN0_BLOCK_W={2,4,8,16}` and sizes the circular buffers from the chosen value. The 32-sample test compares width 4 with width 2 while disabling the new down-projection specialization:

K block	Pure decode mean $\pm$ SD	95% CI	Change
2 tiles	26.122 $\pm$ 0.402 tok/s	[25.977, 26.267]	—
4 tiles	26.192 $\pm$ 0.338 tok/s	[26.071, 26.314]	+0.27%

The intervals overlap and the measured change is small, so width two remains the default. Widths 8 and 16 remain available for experiments. Wider blocks also change accumulation order and can change greedy token sequences.

4.14 110-core fused-residual down projection

The second MLP `matmul` has the exact decode shape 1×12288 by 12288×4096 . The generic program used 64 cores and achieved only 246.2 GB/s of effective BFP8 weight bandwidth, well below the SwiGLU kernel. `down_projection_110_core.patch` adds an exact-shape Blackhole program selected only for BF16 activation/output, BFLOAT8_B weights, a BF16 residual, DRAM-interleaved tensors, and an 11-by-10 worker grid.

The program avoids a K split and cross-core reduction. It partitions the 128 output tile columns across 110 workers:

- 18 wide workers each compute two N tiles, covering columns 0–35;
- 92 narrow workers each compute one N tile, covering columns 36–127;
- one activation sender multicasts to the other 109 workers;
- weights are read exactly once;
- K is processed four tiles at a time with packer-L1 accumulation; and
- the residual addition occurs in the final compute kernel before output.

Wide and narrow workers need different partial-result circular buffers. An early version gave narrow workers a two-tile ring even though they produced one tile. Partial sums alternated slots and corrupted output. Separate one- and two-tile rings fixed the bug.

The exact-shape specialization is enabled by default. `TT_METAL_DOWN_PROJECTION_110_CORES=0` selects the generic program in the same binary for A/B measurement.

4.15 Runtime trace capture for short prefill

Decode replays a recorded TT-Metal command sequence while refreshing request-dependent buffers. Previously, the five-token prompt, padded to one 32-row tile, ran as separate host submissions.

Setting `SGLANG_TT_TRACE_DECODE_ONLY=false` records the fixed-shape prefill sequence too. In a same-binary 32-sample A/B test:

Metric	Decode-only trace	Prefill + decode trace	Change
Time to first token	618.84 $\pm$ 4.76 ms	59.15 $\pm$ 3.70 ms	-90.44% (10.46x)
Total streaming time	5.5067 $\pm$ 0.0598 s	4.9299 $\pm$ 0.0680 s	-10.48%
Pure decode throughput	25.986 $\pm$ 0.315 tok/s	26.079 $\pm$ 0.362 tok/s	+0.36%

Tracing removes about 560 ms before the first token while pure decode changes by +0.36%. In the primary cumulative benchmark, it adds +10.23% end-to-end throughput for a 128-token response.

4.16 Packaging, build, and startup changes

Other patches make the single-library deployment buildable and keep startup cost manageable. They are not decode optimizations.

Table 8: Non-model patch groups in the current libtt build.

Concept	Representative files	Purpose
Embedded runtime assets	<code>tt_metal_runtime_root.cc</code> , <code>tt_metal_runtime_auto_setup.cc</code> , Bazel embed rules	Compress TT-Metal runtime files into <code>libtt.so</code> , extract to a fingerprinted temporary root, and initialize before PJRT.
Static integration and symbol hygiene	<code>BUILD.bazel</code> , <code>libtt_pjrt_exports.lds</code>	Link the upstream plugin and internal libraries into one shared object while exposing only PJRT ABI symbols.
Bazel overlays	<code>third_party/*/BUILD.overlay.bazel</code> , <code>tt_xla.BUILD.bazel</code> , <code>tt_mlir.BUILD.bazel</code>	Bring separately built upstream projects into one central graph with pinned sources and toolchains.
Remove unused fabric/runtime surfaces	<code>disable_fabric*.patch</code> , <code>disable_inspector_runtime.patch</code>	Exclude components unnecessary for the single-chip libtt artifact and avoid unresolved or duplicated runtime dependencies.
Linkability and TLS constraints	<code>reduce_static_tls.patch</code> , <code>remove_initial_exec_tls.patch</code> , public-UMD include patch	Make large statically integrated components safe to load as a PJRT shared library.
Compiler/runtime trimming	TTNN-only registration/runtime patches, cold stubs, disabled TT-Lang resolver, fast sharded-module check	Avoid bringing unused compiler/runtime paths into the artifact and reduce cold compilation work.
API/version compatibility	protobuf, Shardy, affine, constructor-name, map/include patches	Reconcile pinned revisions inside one build without relying on a preinstalled matching stack.

These patches keep the integrated build loadable and remove unused subsystems.

5 Experimental method

5.1 Hardware and software

Item	Value
Accelerator	Tenstorrent Blackhole P150
Firmware observed at startup	19.6.0
Host CPU	Intel Core Ultra 9 185H, 22 logical CPUs online
Host OS/kernel	Linux 6.8.0-124-generic, x86-64
Model	Qwen/Qwen3-8B
Model/activation dtype	BF16
Weight-lowering request	bfp_bf8
Attention backend	TT
Serving frontend	/home/pcmoritz/sglang-jax at 24eb823ed97e
Established main snapshot	627a32d
Current kernel snapshot	37d5460
Benchmark date	15–16 July 2026, America/Los_Angeles

The P150 exposes 120 Tensix workers in this firmware configuration [12]. The SwiGLU program uses 96; the new down projection uses 110. Results are single-device and shape-specific.

5.2 Serving command

We launch the server with the command documented in the repository README:

```
env -u TT_METAL_RUNTIME_ROOT \
  PYTHONPATH=/home/pcmoritz/sglang-jax/python \
  PJRT_NAMES_AND_LIBRARY_PATHS=tt:/home/pcmoritz/libtt/bazel-bin/libtt.so \
  JAX_PLATFORMS=tt \
  JAX_USE_SHARDY_PARTITIONER=false \
  JAX_COMPILATION_CACHE_DIR=/tmp/sglang-jax-qwen3-8b-jax-cache \
  SGLANG_TT_HOST_WEIGHT_LOAD=1 \
  SGLANG_TT_OPTIMIZATION_LEVEL=1 \
  SGLANG_TT_EXPERIMENTAL_WEIGHT_DTYPE=bfp_bf8 \
  SGLANG_TT_TRACE_DECODE_ONLY=false \
  /home/pcmoritz/sglang-jax/.venv/bin/python -m sgl_jax.launch_server \
  --model-path Qwen/Qwen3-8B \
  --host 127.0.0.1 --port 31000 \
  --device tt --dtype bfloat16 --attention-backend tt \
  --max-running-requests 2 --max-total-tokens 1024 \
  --max-prefill-tokens 256 --chunked-prefill-size 256 \
  --page-size 32 --watchdog-timeout 1200 \
  --disable-precompile --skip-server-warmup \
  --disable-overlap-schedule --disable-radix-cache
```

Requests are serial despite the capacity of two. Radix caching and overlap scheduling are disabled so every request executes the same prompt prefill and does not overlap another request. Optional CPU/fused sampling paths remain at their disabled defaults; this report concerns the main model path.

The foundation decomposition uses the same command with `SGLANG_TT_TRACE_DECODE_ONLY=true`. V0 and V1 predate fixed-shape prefill tracing, so this reproduces the timing scope of the original +22.58% measurement. All foundation variants use that setting.

5.3 Request and sample policy

We send the same request 34 times in the primary sequence:

```
{
  "text": "The capital of France is",
  "sampling_params": {"temperature": 0, "max_new_tokens": 128}
}
```

The first two requests are discarded because they include graph compilation, TT-Metal kernel compilation,

trace setup, and cache population. The next 32 complete requests form the analysis window. Primary throughput is

$$T_j = \frac{128}{L_j},$$

where L_j is SGLang’s server-reported end-to-end latency.

For the current kernel experiment, we use the streaming completions endpoint and 32 retained requests per configuration. Pure decode throughput is 127 inter-token intervals divided by the elapsed time from first to last token. Streaming end-to-end throughput is 128 divided by loopback-client total time. The exact per-request observations are in `data/current-kernel-samples.csv`.

5.4 Performance quantities and statistics

For a kernel that appears once in each of the 36 decoder layers, the projected change in token latency is

$$\Delta C = 36 (t_{old} - t_{new}).$$

For the down projection, the measured per-layer difference predicts a 2.241-ms/token reduction; the streaming measurement gives 2.147 ms/token. We compute effective weight bandwidth as $B_{eff} = S_W/t_k$, where S_W is the packed BFP8 weight size and t_k is device kernel time. These quantities let us compare a kernel profile with the end-to-end result without treating the kernel speedup as the model speedup.

Tables report the mean, sample standard deviation, and two-sided 95% Student-t interval. These describe variation within each recorded window but do not remove sequential drift, autocorrelation, or model and shape dependence.

6 Results

6.1 Optimization sequence

V0 is the functional baseline: it runs upstream SGLang-JAX but has few model-specific performance optimizations. V1 through V10 add compiler, runtime, and kernel changes below the XLA/PJRT boundary. Commit IDs are recorded in the CSV files.

Table 10: End-to-end 128-token generation, 32 retained requests per measurement stage.

Stage	Concept introduced	Mean $\pm$ SD (tok/s)	95% CI	Incremental	vs. baseline
V0	Documented serving baseline	16.265 $\pm$ 0.128	[16.218, 16.311]	—	0.00%
V1	Decode foundation: semantic recovery, dtype, and layout	19.938 $\pm$ 0.283	[19.836, 20.040]	+22.58%	+22.58%
V2	Expanded-SiLU recognition	20.596 $\pm$ 0.279	[20.495, 20.696]	+3.30%	+26.63%
V3	QKV projection fusion	20.832 $\pm$ 0.293	[20.726, 20.938]	+1.15%	+28.08%
V4	Two-way shared-LHS matmul	20.479 $\pm$ 0.321	[20.363, 20.594]	-1.69%	+25.91%
V5	Decode RMSNorm width sharding	23.265 $\pm$ 0.235	[23.181, 23.350]	+13.61%	+43.04%

Stage	Concept introduced	Mean $\pm$ SD (tok/s)	95% CI	Incremental	vs. baseline
V6	Consumer-side SiLU/multiply fusion	23.113 $\pm$ 0.296	[23.007, 23.220]	-0.65%	+42.11%
V7	Dst-resident matmul-SwiGLU epilogue	23.744 $\pm$ 0.311	[23.632, 23.856]	+2.73%	+45.98%
V8	Prefill-capable, fallback-free path	23.698 $\pm$ 0.274	[23.599, 23.797]	-0.19%	+45.70%
V9	Fixed-shape prefill trace	26.123 $\pm$ 0.394	[25.981, 26.265]	+10.23%	+60.61%
V10	SwiGLU blocking and 110-core down projection	27.619 $\pm$ 0.322	[27.503, 27.735]	+5.73%	+69.81%

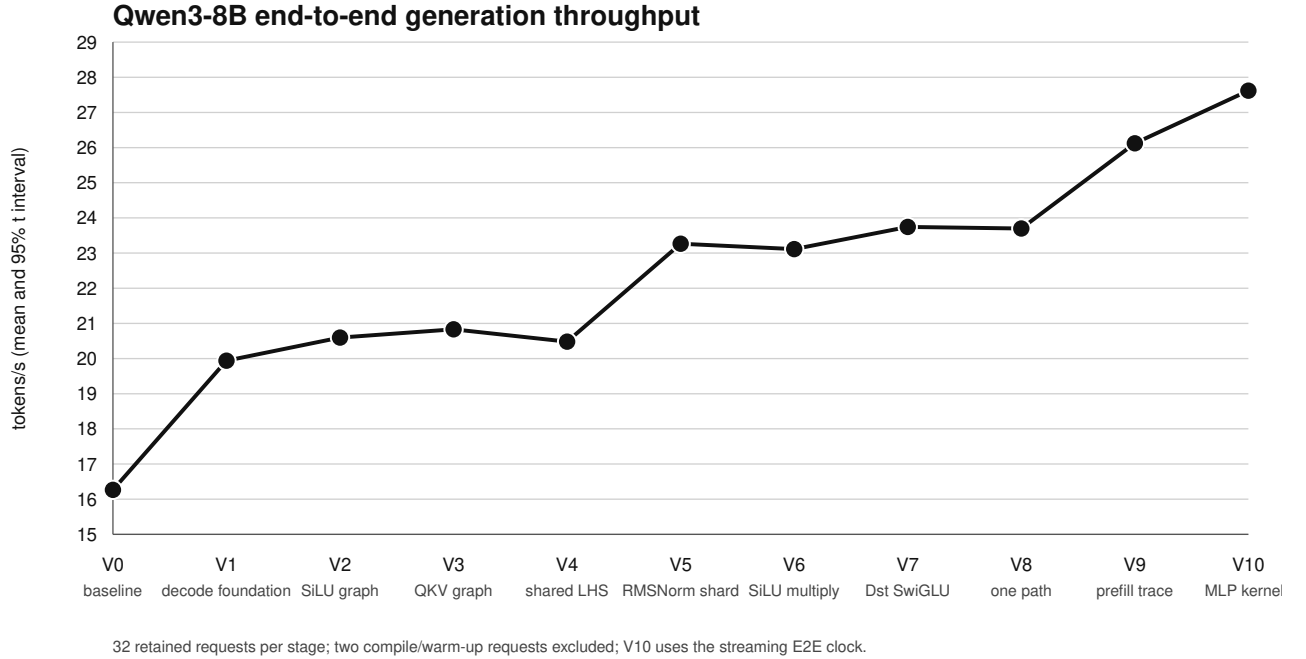


Figure 3: Cumulative throughput across the measured concepts.

The negative and near-zero rows explain the final design: shared-LHS fusion materializes too much data, consumer-side SiLU leaves the producer boundary, and fallback removal changes coverage rather than the kernel.

V0 through V9 use the server-reported request latency from the README benchmark. V10 uses the loopback streaming end-to-end clock needed for the kernel A/B and external-server comparison. The clocks have the same request scope but are collected independently, so the V10 incremental value is the change between stage means, not the isolated kernel effect.

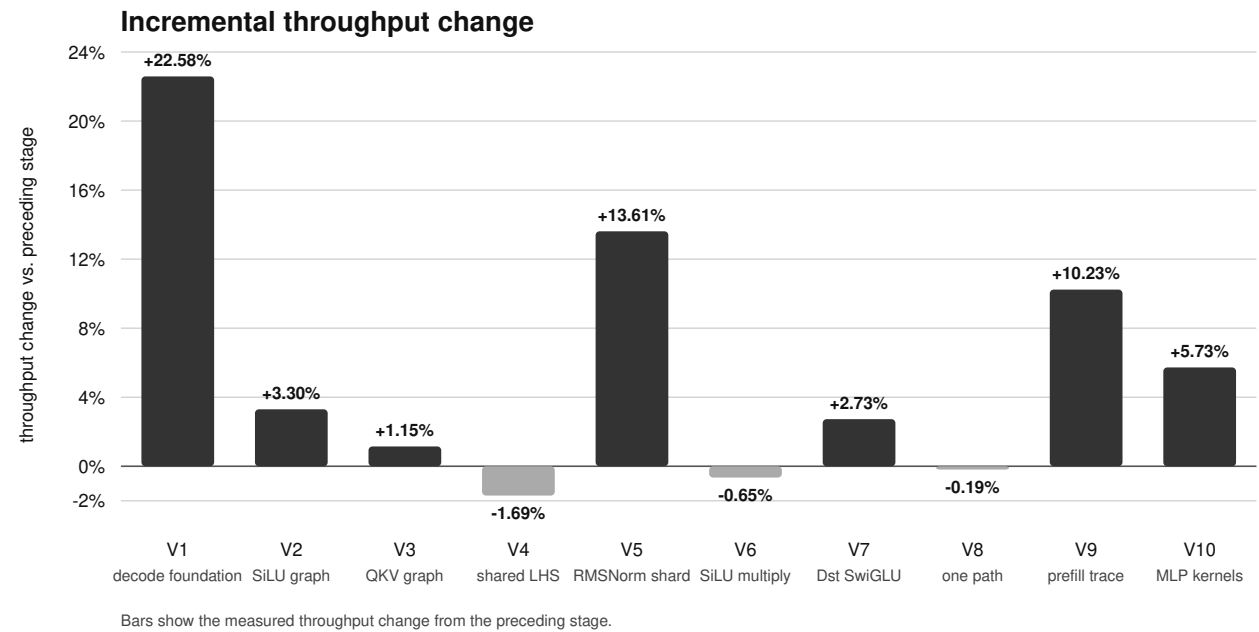


Figure 4: Incremental effect of each concept in the optimization sequence.

6.2 Decomposing the foundation

We repeated the V0/V1 comparison and built six leave-one-concept-out variants from the V1 foundation. The full foundation and functional baseline each have two separate 32-request windows, for 64 retained requests each. Every omission has 32 retained requests. All builds keep the NoC public-include patch because it only fixes the build.

Table 11: Fresh foundation decomposition. Change versus full is the throughput change when the concept is omitted. Marginal speedup is full divided by the omission. Gain share is the omitted throughput loss divided by the total 3.854-token/s foundation gain; it is descriptive and non-additive.

Configuration	N	Mean $\pm$ SD (tok/s)	95% CI	Change vs. full	Marginal speedup	Gain share
Functional baseline	64	16.039 $\pm$ 0.177	[15.995, 16.084]	-19.37%	+24.03%	100.0%
Full foundation	64	19.894 $\pm$ 0.265	[19.827, 19.960]	—	—	—
Without JAX RMSNorm recognition	32	16.678 $\pm$ 0.164	[16.619, 16.737]	-16.16%	+19.28%	83.4%
Without rank-3 decode RoPE fusion	32	19.335 $\pm$ 0.259	[19.242, 19.429]	-2.81%	+2.89%	14.5%
Without KV-cache result typing	32	19.786 $\pm$ 0.289	[19.682, 19.890]	-0.54%	+0.54%	2.8%
Without BF8 activation lowering	32	19.843 $\pm$ 0.255	[19.751, 19.935]	-0.25%	+0.26%	1.3%
Without SiLU call lowering	32	19.983 $\pm$ 0.292	[19.878, 20.088]	+0.45%	-0.45%	-2.3%
Without decode layout admission	32	20.024 $\pm$ 0.234	[19.940, 20.108]	+0.66%	-0.65%	-3.4%

RMSNorm recognition explains most of the gain. Its 3.216-token/s marginal loss is 83.4% of the total baseline-to-foundation difference. Rank-3 RoPE fusion is the second clear contributor at 0.558 tokens/s, or 14.5% of the total. The KV-cache and BF8 activation effects are small. SiLU call lowering and layout admission show no positive decode effect in this graph; they provide graph and layout coverage.

The shares sum to 96.3%, not 100%, because patches interact and each omission is measured with all other foundation patches present. This is not an additive or factorial allocation. The two full-window means are 19.832 and 19.955 tokens/s; the baseline means are 16.043 and 16.036 tokens/s. The fresh total gain is +24.03%. The old +22.58% result used a 16.265-token/s baseline. The new full mean is only 0.22% below the old 19.938-token/s result, while the new baseline is 1.38% lower.

All configurations return 128 tokens and are deterministic within each window. The no-RMSNorm output has a different token hash from the other configurations, although both completions are coherent. This test therefore does not establish equal model quality.

Foundation concept leave-one-out results

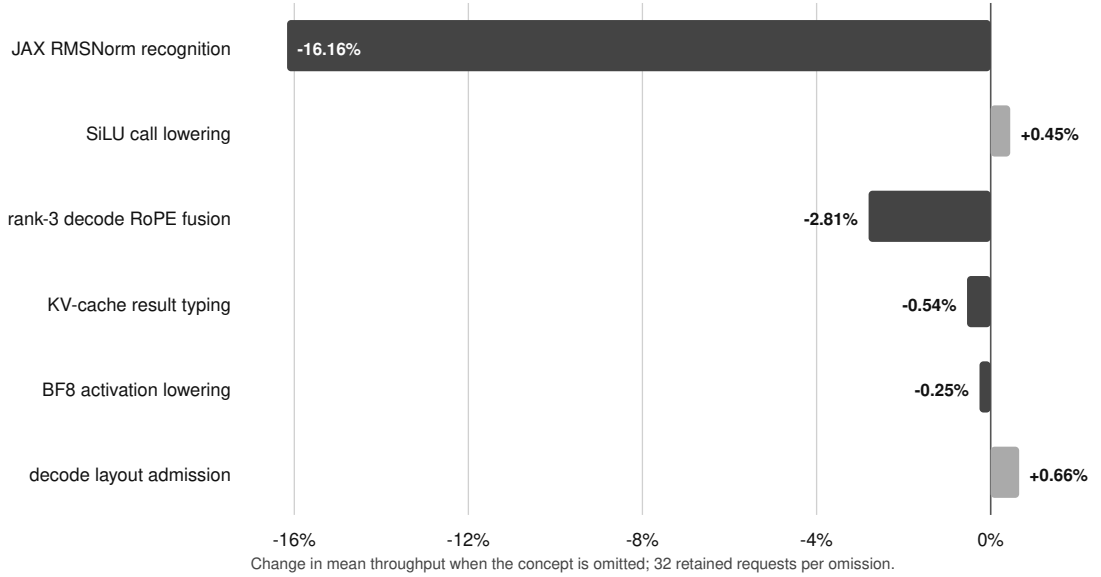


Figure 5: Throughput change when each foundation concept is omitted.

6.3 Isolating the final MLP-kernel stage

The V10 result combines the SwiGLU blocking choice and the 110-core down projection with the preceding compiler, runtime, and kernel optimizations. The same-build measurements below separate their effects.

Table 12: Current-branch streaming results, 32 retained samples per configuration. Width 4 is compared with width 2; the down projection is compared with the same width-4 binary/configuration with the specialization disabled.

Configuration	Pure decode mean $\pm$ SD	95% CI	Streaming E2E mean $\pm$ SD	Change in decode
SwiGLU K block 2, generic down	26.122 $\pm$ 0.402	[25.977, 26.267]	26.016 $\pm$ 0.396	—
SwiGLU K block 4, generic down	26.192 $\pm$ 0.338	[26.071, 26.314]	26.085 $\pm$ 0.333	+0.27%
K block 4, 110-core down	27.753 $\pm$ 0.330	[27.634, 27.872]	27.619 $\pm$ 0.322	+5.96%

The down projection cuts mean decode time from 38.179 to 36.032 ms/token, a 2.147 ms/token saving. Streaming end-to-end throughput rises by 5.88%, from 26.085 to 27.619 tokens/s. TTFT remains about 58 ms.

6.4 Device profile of the down projection

The direct TT-Metal profile identifies 36 occurrences per token, matching the 36 Qwen3-8B decoder layers. Exactly those operation positions change from 64 to 110 workers.

Metric	Generic program	110-core specialization	Change
Workers	64	110	+71.9%
Mean kernel time/layer	217.188 μ s	154.932 μ s	-28.66%
Effective BFP8 weight bandwidth	246.2 GB/s	345.2 GB/s	+40.18%
Projected 36-layer saving	—	2.241 ms/token	—

The projected kernel saving is close to the observed 2.147 ms/token. The new program reaches about 92% of the previously measured 375.1 GB/s SwiGLU effective bandwidth. This agreement between operation profile and end-to-end measurement supports the causal attribution.

6.5 Upstream tt-inference-server reference

Tenstorrent’s tt-inference-server is an OpenAI-compatible service built around vLLM and TT-Transformers [20]. It enters the stack above TTNN rather than through the XLA, StableHLO, and TT-MLIR path:

libtt: SGLang-JAX → JAX/XLA → StableHLO/PJRT → TT-XLA/TT-MLIR → TTNN → TT-Metal

TTIS: OpenAI API → vLLM → TT-Transformers → TTNN → TT-Metal

The two stacks use different porting strategies. TTIS uses TT-specific code in the inference engine, transformer implementation, TTNN, and TT-Metal. libtt keeps the measured SGLang-JAX frontend and JAX model largely unchanged, then optimizes the compiler, runtime, and kernels below the XLA/PJRT boundary.

Both runs use the same P150, Qwen3-8B model, prompt, 128-token output, serial request policy, disabled prefix cache, and persistent loopback collector. Pure decode uses the 127 intervals from first to last token. End-to-end throughput uses request send to last token. Each run has two warm-ups and 32 retained requests.

Implementation	Pure decode mean ± SD [95% CI] (tok/s)	Streaming E2E mean ± SD [95% CI] (tok/s)	TTFT mean ± SD
libtt current (37d5460)	27.753 ± 0.330 [27.634, 27.872]	27.619 ± 0.322 [27.503, 27.735]	58.45 ± 1.50 ms
TTIS v0.10.0	24.887 ± 0.532 [24.695, 25.079]	24.776 ± 0.524 [24.587, 24.964]	63.38 ± 2.10 ms

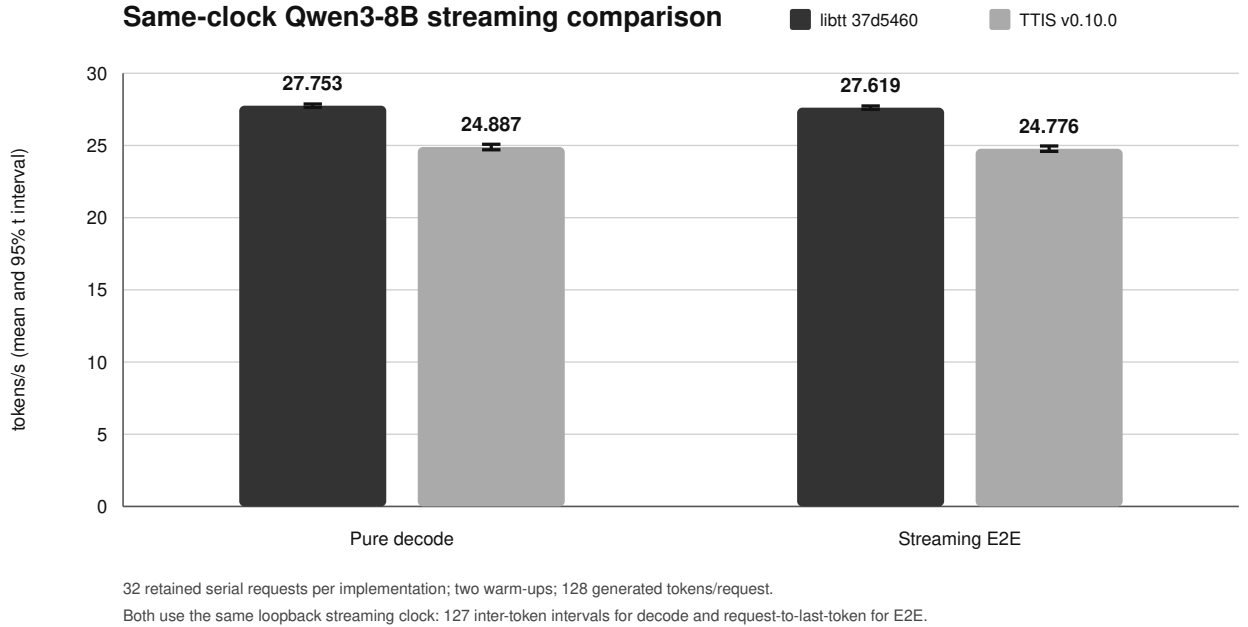


Figure 6: External serving-stack reference on the same model workload.

Under these conditions, libtt is 11.51% faster in pure decode and 11.48% faster end to end. Mean TTFT is 7.78% lower, and mean request-to-last-token latency falls from 5.169 to 4.635 seconds. This compares complete serving stacks, not individual kernels; their model implementations, frontends, and trace strategies differ. The result shows that lower-stack compiler and kernel optimization can match and exceed a stack that also modifies the transformer and inference-engine layers. The tested TTIS artifact is the official v0.10.0 runtime image.

6.6 Correctness and numeric behavior

All 32 requests in each configuration are deterministic. Algebra and reduction changes can alter the token hash. The current kernel hashes are:

Configuration	SHA-256 prefix
SwiGLU block 2, generic down	5119e79e42b5
SwiGLU block 4, generic down	c917933082e3
SwiGLU block 4, 110-core down	c041ccb1901d

Determinism and coherent text do not establish model quality. BF8 arithmetic, reassociation, and reduction order can change logits and greedy choices. Perplexity and task-level evaluation are still required.

7 Discussion

7.1 Placement of transformations

We place each transformation at the highest level that retains the information it needs:

- use TTIR when the problem is algebraic recognition or graph structure;
- use TTNN IR when the compiler needs a named runtime operation, layout, or device dtype;
- use the TTNN runtime when the choice depends on concrete tensor and device state;
- use a TT-Metal program factory for core topology, circular buffers, multicast, and program selection; and
- use device kernels for Dst lifetime, pack/unpack traffic, tile arithmetic, and synchronization.

Lower levels lose graph semantics; higher levels cannot control device data movement.

7.2 Fusion boundaries and intermediate storage

The MLP experiments remove different boundaries:

shared-LHS matmul:

removes duplicate activation work, retains doubled output → -1.69%

SiLU on BinaryNg input:

removes one elementwise intermediate, retains matmul output → -0.65%

matmul-SwiGLU epilogue:

keeps gate/up in Dst and writes only the final half-width result → +2.73%

Fusion should name the Dst, L1, DRAM, or launch boundary it removes.

7.3 Down-projection geometry

Profiling identified weight bandwidth as the down-projection limit. Uneven N partitioning uses 110 workers without rereading weights or reducing partials across cores. It reaches 345.2 GB/s with less complexity than a 2D K/N split.

7.4 Unsuccessful transformations

Wider SwiGLU blocks reduce partial packs but add CB pressure or reduce overlap. The four-tile result changes throughput by only +0.27%, so two remains the default and the larger values remain experimental. Shared-LHS and consumer-side fusion are kept in the record for the same reason: their measured costs explain the final producer-side design.

8 Limitations and future work

1. **One model and decode regime.** Results apply to Qwen3-8B, a five-token prompt padded to one tile, serial 128-token generation, and a single P150. Other batches, contexts, and scheduling policies may have

different limits.

2. **Cumulative attribution is ordered.** The established stages are real cumulative builds, not a factorial experiment. Patch effects can interact.
3. **Foundation effects interact.** The leave-one-concept-out measurements are marginal effects in the complete foundation, not an additive or factorial allocation. Small effects are also close to sequential drift.
4. **Two metric families.** The older cumulative sequence uses server-reported request latency. The current kernel and TTIS comparison uses the same streaming client clock. The two families are not directly interchangeable.
5. **Sequential sampling.** Configurations were not randomized or interleaved. Thermal and background drift can remain.
6. **Profile naming required positional alignment.** The final profiling CSV lacked operation names. Its 854 rows align with an earlier named trace, and the 36 changed positions match the down projections.
7. **Shape-specific kernels.** The 96-core epilogue and 110-core down projection target Blackhole and exact Qwen3-8B decode shapes.
8. **Quality is unmeasured.** Deterministic coherent completion is a smoke test, not evidence of equivalent perplexity or task accuracy.
9. **External reference scope.** The TTIS v0.10.0 image is official, but P150 model metadata was added locally because that release listed Qwen3-8B on P300. The runtime image was unchanged. Both rows use the same collector but different serving and model stacks.
10. **Training is not measured.** The PJRT/StableHLO boundary can carry training graphs, but operation coverage, collectives, optimizer execution, and numerical quality still need implementation and validation.

The next MLP experiment is to keep the SwiGLU output on chip for the down projection. That requires a shared producer/consumer sharding contract.

9 Conclusions

With `libtt.so`, we keep the XLA-facing framework layers close to upstream while packaging the PJRT backend, compiler, runtime, kernels, and runtime assets in one artifact. The report demonstrates that boundary with SGLang-JAX; the same StableHLO interface provides a path to PyTorch through TorchTPU once the required adapter and operation coverage exist. One Bazel graph makes the lower stack available for cross-layer optimization.

We started with a functional backend that runs upstream SGLang-JAX at 16.265 tokens/s. Compiler, runtime, and kernel changes raise measured end-to-end throughput by 69.81% to 27.619 tokens/s and deliver 27.753 decode tokens/s. Under the matched benchmark conditions, libtt is 11.51% faster than the TTIS decode mean even though TTIS also uses TT-specific transformer and inference-engine layers.

We measured JAX inference. The PJRT/StableHLO architecture is not tied to JAX or inference and can support JAX or PyTorch training once the required operations, collectives, optimizer graphs, frontend adapter, and validation are added.

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